Revision Table

Revision No.	Details	Section / Page No.
1	Initial thesis structure setup	All chapters
2	Added LaTeX packages for code and formatting	Preamble
3	Restructured to match approved ClassiFish format	Chapter organization
4	[Future revisions to be added here]	[Location]

Table 1: Thesis Revision History

Consol: A Notes-to-Recollection Application Using SimCSE for its Textual Semantic Comparison

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Abstract

Generative AI has presented society with an opportunity to create a surplus of educational content in mere seconds. While knowledge is inherently beneficial and promotes intellectual and academic growth, there still is a bridge to actual learning that cannot be crossed just through using AI and its instantaneous generative abilities (Mishra & Anthony, 2023). Learning, in the academic landscape, is comprised of obtaining information which can be in the form of notes (Rashid & Rigas, 2006; Devlin et al., 2019). Retaining knowledge after note-taking remains a challenge, as forgetting is inevitable and overlooked by recent AI and NLP advancements. (Klemm, 2012; Naim et al., 2020). In studying and reviewing, if a person wishes to ensure their recollection of key topics is accurate and coherent, options are limited without manually referencing source material, which can be tiring and inefficient. Assistance is needed, particularly in providing instant feedback for accuracy and mastery of notes (Karpicke & Roediger, 2008; Sachs, 1967). This research proposes Consol, a notes-to-recollection application that uses SimCSE's contrastive sentence embeddings and semantic similarity analysis to compare how similar a person's recollection is to their original notes (Gao et al., 2021; Corley et al., 2005), and uses progress tracking techniques to encourage consistency and improvement (Deterding et al., 2011; Gaston & Cooper, 2017). It will be a web-based application that leverages semantic similarity evaluation to assess memory recall effectively. The study aims to evaluate whether semantic-based feedback enhances learning outcomes compared to traditional recall methods. Through system testing and user performance assessment, this research will analyze the system's usability, accuracy, and effectiveness in improving memory retention (Baars et al., 2022; Tran & Nguyen, 2022). The results aim to provide a scalable solution bridging note-taking and active learning while fostering consistent knowledge consolidation (Vișcu, 2024; Stojanovic et al., 2020).

Contents

List of Figures

List of Tables

1

Introduction

Memory is not merely the result of recording information but the outcome of consistent review and active engagement with what has been learned. Notes, while essential for capturing ideas, do not inherently lead to long-term retention or understanding simply by being created. Instead, it is through revisiting, processing, and consolidating these notes that individuals truly reinforce their knowledge (Karpicke and Roediger, 2008; Klemm, 2012).

For students, teachers, and professionals alike, the challenge lies in establishing consistent learning practices and receiving validation for their efforts—validation that not only measures recall but also provides meaningful feedback on comprehension (Stojanovic et al., 2020; Baars et al., 2022). Traditional note-taking methods often focus on transcription, prioritizing the exact wording of information over its deeper meaning. However, research by Sachs (1967) highlights that individuals naturally retain the semantic meaning—the "gist" of information—rather than its verbatim form.

This tendency to paraphrase and restructure ideas reflects the cognitive process of abstraction, where understanding takes precedence over memorization. Conventional systems that evaluate recall by word-for-word accuracy fail to account for this natural process, offering an incomplete and often discouraging measure of learning performance (Corley et al., 2005; Gao et al., 2021).

Additionally, progress tracking has proven to be an effective strategy for sustaining motivation and fostering consistent learning habits. By incorporating elements such as performance metrics, mastery indicators, and progress tracking, such systems provide users with actionable insights and a sense of achievement (Deterding et al., 2011; Gaston and Cooper, 2017). These tools not only validate the user's learning progress but also make the process more interactive and rewarding, promoting deeper engagement and long-term retention (Tran and Nguyen, 2022; Viscu, 2024).

1.1 | Background of the Study

Now, as technology continues to advance in the current "information age," the demand for fast, accurate, and easily accessible information has increased dramatically. This is particularly evident in fields where knowledge acquisition is critical, such as education, research, and personal development. The proliferation of generative AI technologies and digital tools has revolutionized the ways individuals gather, process, and manage information. Digital note-taking tools, such as Evernote, Notion, Obsidian, and various AI-assisted applications, offer features like automatic transcription, cloud synchronization, and summarization, making the recording of ideas and knowledge more efficient and convenient to the modern user (Baars et al., 2022; Rashid and Rigas, 2006).

Existing studies confirm that effective learning requires more than passive note creation. Research conducted by Sachs (1967) indicates that individuals are better at retaining semantic meaning rather than exact syntactic details. This means that during memory retrieval, individuals often paraphrase, restructure, and abstract ideas rather than reproduce content verbatim. Traditional systems that rely on word-for-word accuracy to evaluate recall fail to account for this natural cognitive process, leading to an incomplete understanding of memory performance (Corley et al., 2005; Gao et al., 2021).

Building on this, the integration of AI-driven systems in learning environments has opened new avenues for assessing memory and comprehension more effectively. By including the more recent advancements in natural language processing (NLP) and similarity-based algorithms, these systems can evaluate recall beyond mere verbatim reproduction, focusing instead on the semantic alignment between original and recalled content (Devlin et al., 2019; Gao et al., 2021). Such approaches recognize the inherent flexibility of human cognition, where meaningful understanding takes precedence over exact replication. For example, sentence-transformer models like SimCSE have demonstrated the ability to measure semantic similarity with high accuracy, making them suitable for applications that assess memory recall in a way that mirrors human evaluation (Gao et al., 2021; Corley et al., 2005).

This shift toward semantic-based assessment not only provides a more accurate measure of knowledge retention but also aligns with modern pedagogical principles that emphasize deep learning and critical thinking. By rewarding comprehension and conceptual mastery over rote memorization, AI-assisted systems encourage learners to engage more actively with the material, fostering long-term knowledge retention and improved cognitive performance (Karpicke and Roediger, 2008; Klemm, 2012). As these technologies continue to evolve, they hold the potential to transform traditional methods of learning evaluation and pave the way for more adaptive and personalized educational tools (Stojanovic et al., 2020; Viscu, 2024).

1.2 | Statement of the Problem

The rapid development and advancement of Generative AI has led to the overwhelming abundance of knowledge, and with that, an unlimited opportunity to take notes. While this is clearly for some benefit, it is not put into actual use if the increasing number of notes are not consolidated into memory, thereby hindering the cumulative learning process (Karpicke & Roediger, 2008; Klemm, 2012). Additionally, in individual study contexts, there is a lack of mechanisms to verify whether a learner has truly understood or mastered the material. This problem is compounded by the fact that human recall often involves paraphrasing and abstracting information, making it difficult for current systems to accurately assess a learner's understanding (Sachs, 1967; Corley et al., 2005). Consequently, there is a need for a system that can evaluate learning by recognizing and comparing both the original content and its reconstructed ideas through context clues, ensuring that abstracted yet relevant concepts are acknowledged without overlooking their presence in the material (Gao et al., 2021; Devlin et al., 2019).

1.3 | Objectives

1.3.1 | General Objective

To develop a web-based memorization system that refers to the user's pre-made notes and prompts the user to perform active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas.

1.3.2 | Specific Objectives

1. Design the system architecture of the notes-to-recollection comparison system, including its scoring criteria for evaluating recall accuracy.
2. Devise pre-processing procedures for removing unnecessary non-alphanumeric symbols and formatting artifacts to be suitable for semantic comparison.
3. Employ Simple Contrastive Sentence Embeddings (SimCSE) for semantic comparison of two given texts.
4. Implement a web-based application integrating the SimCSE Model, giving it an appropriate user experience that facilitates motivation through progress tracking and active learning.
5. Conduct a usability test on the web-based application and document the results.

1.4 | Scope and Limitation

The study is strictly limited to a maximum of 10 test participants. The study will be conducted from January 2025 through April of the same year.

1.5 | Significance and Contribution

The proposed implementation will assist users in memorizing their notes and incentivize steady progress in content mastery through various tracking strategies.

2

Review of Related Literature

This chapter reviews relevant literature on note-taking strategies, memory processes, natural language processing (NLP) for semantic similarity, and progress tracking techniques. These areas provide the theoretical basis for understanding information encoding, memory consolidation, textual comparison, and user engagement. By examining these interconnected concepts, the review establishes a foundation for exploring methods that enhance learning, recall, and performance. The studies discussed offer insights into optimizing knowledge retention, evaluating recall accuracy, and motivating users through structured feedback mechanisms.

2.1 | Memory and Cognitive Processes

Memory plays a crucial role in the learning process, specifically in the retention and recall of information. Studies have demonstrated that memory prioritizes the retention of meaning over surface-level details, which has direct implications for comprehension and learning.

2.1.1 | The Role of Semantic and Syntactic Memory

Sachs (1967) explored the relationship between semantic and syntactic memory in the context of comprehension. In her study, participants were presented with sentences and tested on their ability to recognize changes in meaning and form after brief intervals. The findings revealed that memory retention is primarily focused on semantic content, while syntactic details, such as active-to-passive transformations, fade quickly. Even when participants retained the overall meaning of the sentences, the formal properties of the sentences were quickly forgotten. This study concludes that memory is structured to prioritize the "deep structure" or meaning of a sentence over its "surface structure" or syntax, suggesting that comprehension relies on extracting and retaining the core meaning of information.

2.1.2 | Cognitive Processes in Memory Retention

Rashid and Rigas (2006) provide an analysis of memory stages and their relationship to learning and information retention. The study outlines the progression of information through sensory memory, short-term memory, and long-term memory, highlighting the necessity of active cognitive processing for effective retention. Encoding, which involves the initial capture of relevant information, and reviewing, which reinforces the transfer of information to long-term memory, are identified as critical phases in the learning process. This study also discusses retrieval failure as a cause of forgetting, where proactive interference and retroactive interference disrupt recall.

Proactive interference occurs when previously learned information impairs the learning of new material, while retroactive interference arises when newly acquired knowledge interferes with the recall of older information. These findings emphasize that memory retention is optimized when learners engage in deep processing activities, such as organizing and summarizing information, rather than relying on surface-level strategies. This perspective aligns with the findings of Sachs (1967), reinforcing the importance of meaningful encoding and the prioritization of core content over superficial details.

2.1.3 | Memory and Recall

Naim et al. (2020) identified that recall performance grows sublinearly as the list length increases. This sublinear growth highlights a saturation effect, where the number of items retrieved plateaus despite an increase in encoded material. The study also revealed that encoding quality plays a crucial role in determining recall performance. Slower presentation rates allowed participants to encode a higher number of words into memory, which improved the total recall capacity. These findings provide evidence that memory recall adheres to universal laws, offering a structured and deterministic perspective on retrieval processes.

2.1.3.1 | Retrieval-Based Testing

Karpicke and Roediger (2008) emphasize that testing is not merely a means of assessment but a potent strategy for enhancing long-term memory retention. Their study debunks the conventional belief that once information is initially learned, further practice adds little value. Instead, repeated retrieval—through testing—promotes deeper encoding and strengthens memory consolidation, highlighting retrieval as an active mechanism essential for sustained learning performance.

2.1.3.2 | Optimizing Retrieval Practice for Long-term Retention

Karpicke and Roediger (2008) demonstrated that retrieval practice plays a central role in consolidating memories and improving long-term retention. Their experiments revealed that repeated retrieval significantly outperforms repeated study in enhancing memory performance. This finding validates the importance of active recall over passive encoding strategies.

2.2 | Note-Taking Techniques and Tools

Note-taking serves as an essential tool for facilitating memory retention and comprehension. Various techniques and tools have been developed to assist learners in organizing and processing information effectively.

2.2.1 | Traditional Note-Taking Techniques

Rashid and Rigas (2006) examine the comparative effectiveness of traditional note-taking methods, including the Cornell Method, the Outlining Method, and the Mapping Method. These methods provide learners with strategies for organizing content to improve comprehension and retention.

2.2.2 | Digital Note-Taking and E-Learning Tools

Rashid and Rigas (2006) further explore the role of technology in enhancing note-taking and memory retention within e-learning environments. The integration of tools such as Microsoft OneNote and Tablet PCs has provided learners with new ways to engage with content, enabling multimedia annotations and portable accessibility.

2.2.2.1 | Quizlet

Tran and Nguyen (2022) demonstrate that Quizlet integrates notes, spaced repetition, and active learning into an engaging platform, effectively improving vocabulary retention and long-term memory consolidation.

2.2.2.2 | ChatGPT and the Generation of Educational Content

Mishra and Anthony (2023) highlight both the benefits and drawbacks of AI tools like ChatGPT in education, emphasizing the need for responsible integration to enhance critical thinking and active learning processes.

2.2.2.3 | Self-Regulated Learning by Using Study Applications

Baars, Khare, and Ridderstap (2022) demonstrate that study applications with well-designed user interfaces support self-regulated learning by enabling goal-setting, time management, and feedback during study sessions.

2.3 | NLP and Semantic Similarity

The development of NLP techniques has significantly advanced models that analyze and quantify semantic similarity.

2.3.1 | Text Semantic Similarity

Corley et al. (2005) provided foundational insights into computational methods for evaluating semantic similarity, focusing on techniques that measure the closeness of meaning between texts.

2.3.2 | The BERT Framework

Devlin et al. (2019) emphasize that BERT leverages bidirectional representations of language to capture semantic relationships through its transformer-based architecture.

2.3.2.1 | SimCSE

Gao, Yao, and Chen (2021) developed SimCSE, a fine-tuning approach that enhances the quality of sentence embeddings, making them suitable for tasks like semantic similarity evaluation and textual entailment.

2.3.2.2 | Contrastive Learning

SimCSE's unsupervised framework generates "positive pairs" by applying independent dropout masks to the same input sentence. Specifically, when a sentence is passed through the BERT encoder twice, the randomness introduced by dropout produces two slightly different embeddings.

$$\ell_i = -\log \frac{\exp(\mathrm{sim}(h_i, h_i^+)/\tau)}{\sum_{j=1}^{N} \exp(\mathrm{sim}(h_i, h_j)/\tau)}$$

Figure 2.1: Formula for Contrastive Learning

2.3.2.3 | Cosine Similarity

Cosine similarity quantifies the angular distance between embeddings, ensuring that semantically similar sentences are pulled closer together, while dissimilar ones are pushed apart. The temperature parameter τ sharpens the similarity distribution, controlling the influence of positive and negative pairs during training.

$$\mathrm{sim}(h_1, h_2) = \frac{h_1^\top h_2}{||h_1|| \, ||h_2||}$$

Figure 2.2: Formula for Cosine Similarity

2.4 | Progress Tracking

Progress tracking refers to systems that enable users to monitor and evaluate their advancement toward specific goals.

2.4.1 | Progress Tracking as Part of Gameful Design

Deterding et al. (2011) emphasize the importance of structured systems that guide users through progressive goals and stages of development, helping maintain motivation and engagement.

2.4.1.1 | The Three-Star Scoring System

Gastón and Cooper (2017) explore the efficacy of the three-star scoring system, highlighting its ability to improve user engagement and performance in educational and gamified applications.

2.5 | Summary

The studies reviewed in this chapter highlight key advancements in note-taking strategies, memory processes, NLP, and progress tracking. These insights establish a foundation for the proposed system, integrating effective learning strategies and semantic similarity evaluation to optimize recall and knowledge retention.

The table compares various studies based on their focus on specific aspects or features that are relevant to the proposed approach. It highlights which aspects are addressed and identifies gaps that the proposed approach aims to fill by integrating these features into a unified framework.

Aspects/Features: Studies:	Integration of Semantic Similarity Models	Progress Tracking	Focus on Knowledge Consolidation, Memory, and Recall	Flexibility for Varied Note Structures (Formatting)	Feedback and Performance Metrics	App with Interactive UI for Enhanced Usability	Use of NLP for Evaluating Semantic Understanding
Stojanovic et al., 2020	✗	✗	✗	✗	✗	✅	✗
Tran et al., 2020	✗	✅	✅	✗	✅	✅	✗
Baars et al., 2003	✗	✗	✅	✗	✗	✅	✗
Proposed Approach	✅	✅	✅	✅	✅	✅	✅

Figure 2.3: Comparison Table of Related Literature

3

Theoretical Framework

4

Methodology

4.1 | Research Methodology

4.2 | System Design Methodology

4.3 | Development Process

4.4 | Data Collection Methods

4.5 | Evaluation Framework

4.5.1 | SimCSE Similarity Threshold Calibration

One of the critical aspects of developing Consol was establishing appropriate similarity thresholds for the star rating system that would provide meaningful and fair assessment of student recall performance. This required systematic validation of SimCSE's behavior across various linguistic scenarios commonly encountered in educational contexts.

The initial threshold hypothesis was established based on educational assessment principles:

- 3 stars (Excellent): Similarity $\geq$ 0.81 - Demonstrates comprehensive understanding
- 2 stars (Good): Similarity $\geq$ 0.60 - Shows adequate recall with minor gaps

- 1 star (Basic): Similarity ≥ 0.30 - Indicates partial understanding
- 0 stars (Poor): Similarity < 0.30 - Requires significant improvement

To validate these thresholds, we designed a comprehensive test suite encompassing various linguistic transformations that students might employ during recall sessions. The test cases were selected to represent common scenarios in educational recall, including paraphrasing, summarization, expansion, and potential edge cases that could lead to misleading similarity scores.

4.5.2 | Test Case Design and Implementation

Seven representative test cases were developed and evaluated to assess SimCSE performance across different linguistic scenarios. Each test case consisted of an original sentence (S1) and a student response variant (S2), with expected similarity ranges based on semantic preservation and educational value.

Figure **??** presents the complete set of test cases used for threshold validation, while Table **??** summarizes the quantitative results and analysis.

The test cases were specifically designed to address:

1. **Semantic Opposition**: Testing SimCSE's ability to detect contradictory meanings
2. **Paraphrasing Detection**: Evaluating performance on synonym-based rewording
3. **Content Expansion**: Assessing similarity when students provide additional detail
4. **Content Summarization**: Testing condensed versions of original content
5. **False Similarity Traps**: Identifying cases where word overlap doesn't indicate understanding
6. **Structural Variations**: Examining different sentence constructions with preserved meaning
7. **Formatting Robustness**: Testing resilience to punctuation and formatting changes

This systematic approach ensured that the final threshold settings would be both educationally meaningful and technically robust, providing students with fair and consistent assessment of their recall performance.

1. **Simple Opposite Meanings**
   - S1: *The apple is green.*
   - S2: *The apple is red.*
   - Expected Score: 0.40 - 0.60
2. **Paraphrased Sentences (Synonyms Rewording)**
   - S1: *She quickly finished her homework.*
   - S2: *She completed her assignment in no time.*
   - Expected Score: 0.75 - 0.95
3. **Sentence Expansion (Short to Long Text)**
   - S1: *The cat sat on the mat.*
   - S2: *A small, fluffy cat was resting on the comfortable mat by the fireplace.*
   - Expected Score: 0.65 - 0.85
4. **Summarization (Long to Short Text)**
   - S1: *The city of Rome, known for its rich history and cultural heritage, is home to iconic landmarks such as the Colosseum and Vatican City, attracting millions of tourists every year.*
   - S2: *Rome is a historic city with famous landmarks and many visitors.*
   - Expected Score: 0.70 - 0.85
5. **Common Words, Different Meaning (False Similarity Trap)**
   - S1: *She runs a software company.*
   - S2: *He runs five miles every morning.*
   - Expected Score: 0.30 - 0.50
6. **Different Sentence Structures (Same Idea, Different Style)**
   - S1: *The storm caused massive destruction in the city.*
   - S2: *A devastating storm wrecked the city.*
   - Expected Score: 0.75 - 0.95
7. **Adding Markdown & Symbols (Effect on Similarity)**
   - S1: *The quick brown fox jumps over the lazy dog.*
   - S2: *The quick brown fox jumps over the lazy dog!*
   - Expected Score: 0.85 - 0.98

Figure 4.1: SimCSE Test Cases for Threshold Validation: Seven representative scenarios designed to evaluate semantic similarity detection across common linguistic variations in educational contexts.

Table 4.1: SimCSE Threshold Validation Results: Comparison of expected vs. actual similarity scores across seven test scenarios

Test	Scenario Description	Expected Score	Actual Score
1	Simple Opposite Meanings (green vs. red apple)	0.40 - 0.60	0.8988
2	Paraphrased Sentences (homework completion)	0.75 - 0.95	0.6628
3	Sentence Expansion (cat on mat elaboration)	0.65 - 0.85	0.8407
4	Summarization (Rome city description condensed)	0.70 - 0.85	0.8295
5	False Similarity Trap (different meanings of "runs")	0.30 - 0.50	0.2707
6	Different Sentence Structures (storm destruction)	0.75 - 0.95	0.8846
7	Markdown & Symbols (punctuation effects)	0.85 - 0.98	0.9750

Note: Expected scores represent anticipated similarity ranges based on semantic preservation and educational assessment principles. Actual scores reflect SimCSE model output during systematic testing phase.

4.5.3 | Validation Results and Threshold Refinement

The systematic testing revealed both strengths and areas for improvement in SimCSE's performance for educational assessment. Analysis of the seven test cases provided crucial insights into the model's behavior across different linguistic scenarios.

Key Findings:

- **False Similarity Detection**: SimCSE successfully identified semantically different sentences with common words (Test 5: 0.27 score), validating its ability to prevent false positive assessments.
- **Structural Robustness**: The model demonstrated strong performance with different sentence structures expressing the same concept (Test 6: 0.88 score), indicating reliable semantic understanding.
- **Formatting Resilience**: Punctuation and formatting changes showed minimal impact on similarity scores (Test 7: 0.98 score), ensuring consistent assessment regardless of student writing style.
- **Paraphrasing Sensitivity**: Synonym-based paraphrasing yielded lower scores than anticipated (Test 2: 0.66 vs expected 0.75-0.95), suggesting the need for

threshold adjustment to accommodate valid rewording.

- **Opposite Meaning Handling**: Contrary to expectations, semantically opposite statements received high similarity scores (Test 1: 0.90 vs expected 0.40-0.60), indicating a limitation in contradiction detection.

These findings informed the refinement of similarity thresholds and provided validation that SimCSE could serve as a reliable foundation for educational assessment, while highlighting specific scenarios requiring careful interpretation in the final system implementation.

4.6 | Implementation Tools and Technologies

4.7 | Testing and Validation Approach

5

Results and Discussion

5.1 | System Implementation Results

5.2 | Performance Analysis

5.3 | Similarity Scoring Validation Results

5.4 | User Experience Testing Results

5.5 | Learning Analytics Dashboard Results

5.6 | Discussion of Findings

5.7 | Limitations and Challenges

6

Conclusions

This section should have a summary of the whole project. The original aims and objective and whether these have been met should be discussed. It should include a section with a critique and a list of limitations of your proposed solutions. Future work should be described, and this should not be marginal or silly (e.g. add machine learning models). It is always good to end on a positive note (i.e. 'Final Remarks').

6.1 | Achieved Objectives

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.2 | Critique and Limitations

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.3 | Future Work

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.4 | Final Remarks

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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